

Gerardo Toboso

Data & Cloud Engineer | AWS Serverless Architect | Cost-Optimized Systems

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Summary

Data & Cloud Engineer who builds production-grade AWS serverless systems and diagnoses failure at scale. At a US mortgage fintech I cut 22M backend calls/week, secured 71M records, and audited 268 production incidents across 12 integrations, identifying 27 systemic root causes and implementing fixes that eliminated 160+ support escalations per cycle.

Skills

Cloud Architecture & Serverless Systems: AWS Lambda, DynamoDB, SQS, API Gateway, WebSockets, Event-Driven design, IAM, and cost optimization

Data Engineering, SQL & Processing: Python (pandas, PySpark, duckdb), PostgreSQL, ETL/ELT pipeline development, Airflow, idempotent design, Parquet partitioning, and data contracts

Reliability & Observability: CloudWatch, structured logging, alerting, automated testing (Pytest), freshness checks, and failure detection

Experience

Fullstack Software Engineer | Cloud & Serverless Architect, Figure (US-based Mortgage Platform) – Contractor · Buenos Aires Province, Argentina · Remote Jan 2024 – present

- Cut 22M redundant backend calls per week by fixing a high-frequency polling inefficiency and reducing AWS Lambda and DynamoDB cost.
- Led a security audit to redact OAuth and API key credentials from 71M historical records in DynamoDB and S3, then added automated redaction in the communication wrapper.
- Migrated a polling-based frontend to an event-driven architecture using API Gateway WebSockets and DynamoDB Streams, enabling sub-2s updates for critical loan state changes.
- Built and maintained a mortgage pipeline orchestrating 50+ AWS Lambdas and 30+ external APIs, reducing loan closing times from 30 days to under 7 and serving 200+ clients with 99.9% availability.
- Conducted a structured root-cause audit of 268 production incidents across 12 third-party integrations (Plaid, Docutech, Persona, Service Link) over 8.5 weeks, identified 27 distinct failure patterns, and implemented targeted fixes that eliminated over 160 manual support escalations per 8-week cycle.
- Resolved a critical Plaid income-validation defect by replacing exact string matching with Unicode normalization and Jaro-Winkler fuzzy matching, eliminating 57 support tickets per cycle, the highest-frequency failure class on the platform, with zero manual ops intervention required after the fix.
- Implemented structured error classification and exponential backoff retry for the Docutech document-generation integration, distinguishing auth failures, payload errors, and service outages, converting 35 recurring opaque API failures per cycle into self-diagnosable, auto-recoverable incidents.
- Fixed a Persona identity-verification defect that treated transient API timeouts (canceled status) as genuine identity failures, generating 22 false rejections per cycle; implemented auto-retry for canceled checks and Unicode normalization on name fields, eliminating all false rejections.
- Resolved a race condition in the Service Link title-order flow where Lambda timeouts between API creation and DynamoDB persistence permanently lost order IDs, replaced dual writes with DynamoDB transactions and SQS-backed retry, eliminating cascading failures across 3 downstream webhook handlers.

Education

Universidad Nacional de San Martín (UNSAM), BSc in Data Science

July 2022 – present

- 75% completed — GPA: 9.0 / 10